A swarm that owns a proof can still lose it

Simulation 1, setup `task003-symmetric`: first results

LLM-free simulator; extended with the efficiency of control, 12 October 2026; runs of 7 October 2026

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact about it and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. An episode has 30 rounds, each a day and a night. During the day the agents act one at a time in a new random order: an agent reads q posts drawn at random from a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* (*argmax*: the most probable allocation; *probability matching*: a vote drawn from its belief) and posts one remembered fact that supports its vote, or nothing if none does. At night a *controller* with a target, A0 (truth) or A2 (false), reads q_c agent posts; unless at least 75% of them come from agents voting for its target, or what it read already proves its target, it posts the b facts from its own pool that best support its target. The board is then cleared, keeping only the controller's new posts. At each dawn every remembered fact is forgotten with probability $1 - \rho$ and comes back if read again. Every episode is run three times with the same random draws: without a controller, with the truth controller and with the false controller.

Summary. 24 agents together hold a complete proof that allocation A0 is correct, but no single agent does. A controller steers toward A0 (truth) or A2 (false). (i) **Whether the false controller wins depends on memory**: when agents forget, it overturns the proof; with full memory its success is temporary, because its own true facts end up completing proofs of A0. (ii) **The controller senses the swarm accurately**, so its failures are not sensing failures. (iii) **Helping the truth is not the same as helping agents assemble their own proof**: extra truth posts raise the share of agents holding a proof, but lower the share holding one from their own evidence.

1 Setup

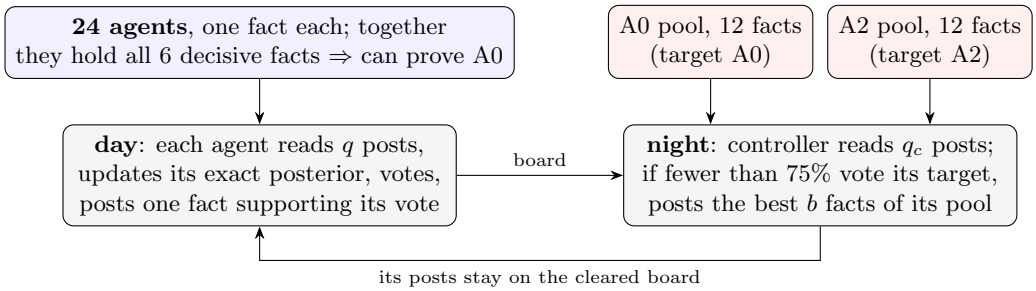


Diagram 1. One round of Simulation 1: a day followed by a night, repeated for 30 rounds. Each dawn, every remembered fact survives with probability ρ.

The two pools share no fact, contain no decisive fact, cannot prove their target, and are matched in strength pair by pair. The agents start, in expectation, with 9 / 7 / 8 votes for A0 / A1 / A2. Agents are exact Bayesian reasoners: an agent's belief is the exact posterior over the three allocations given the facts it remembers. Three agent rules are compared: *argmax* voting with posts that support the vote (the *base* rule), *probability matching* (vote drawn from the posterior), and argmax with *uniform* posts (a random remembered fact, whatever the vote).

Measuring control. Every episode is run three times with the same random draws: silent, truth controller, false controller. The *paired gain* of a controller in the share of agents voting for allocation a on day t is

$$G_t^{(a)} = \frac{1}{E} \sum_{e=1}^{E} \Big[x_{a,t}^{\text{ctrl}}(e) - x_{a,t}^{\text{silent}}(e) \Big], \tag{1}$$

with $E = 1000$ episodes per cell and 95% intervals from resampling episodes.

2 A swarm that owns a proof can still lose it

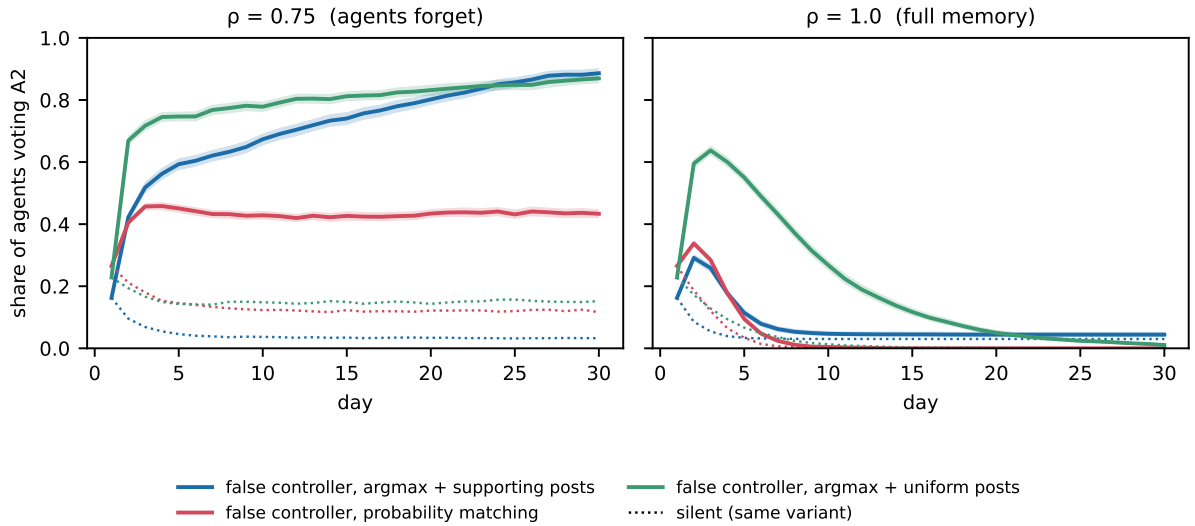


Figure 1: Share of agents voting A2 over the days, under the false controller (solid, 95% band) and without a controller (dotted). $q = 6$, $q_c = 12$, $b = 3$.

Figure 1 shows the central result. **When agents forget** ($\rho = 0.75$), the false controller pushes the A2 share from under 0.2 to about 0.9 under the base and uniform rules, and to about 0.45 under probability matching. **With full memory** ($\rho = 1$) every rule shows the same shape: an early A2 peak (up to 0.6 under uniform posts), then a return to A0 by day 20. Table 1 gives the day-30 gains.

Table 1: Day-30 paired gain of the false controller in the A2 share, Eq. (1); $q_c = 12$, $b = 3$.

ρ	agent rule	$q = 3$	$q = 6$	$q = 12$
0.75	argmax + supporting posts	0.95	0.85	0.26
0.75	probability matching	0.48	0.32	0.08
0.75	argmax + uniform posts	0.67	0.72	0.69
1.0	argmax + supporting posts	0.03	0.01	0.00
1.0	probability matching	0.00	0.00	0.00
1.0	argmax + uniform posts	0.05	0.01	0.01

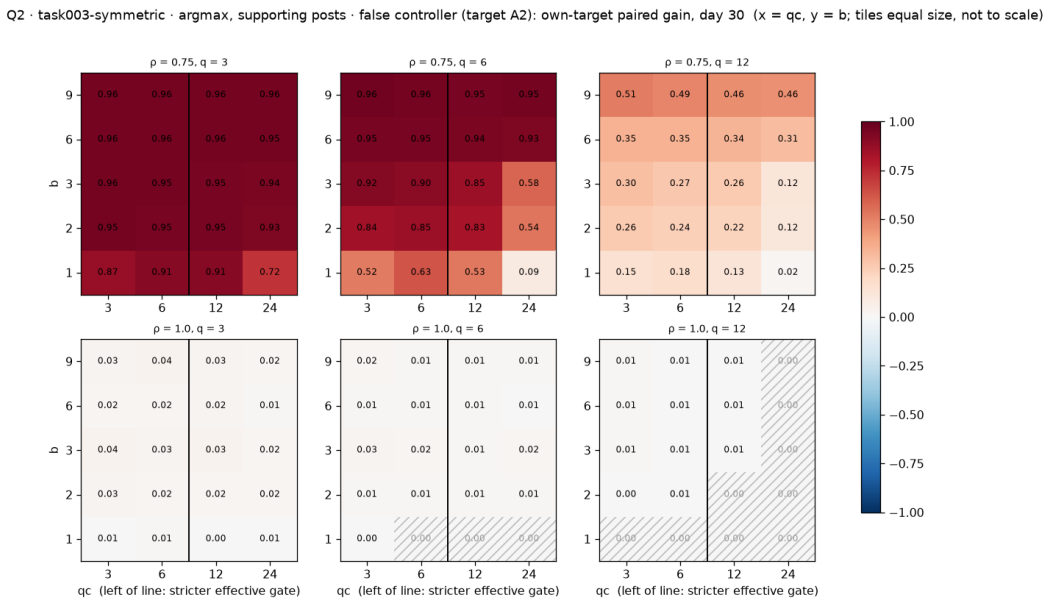


Figure 2: Day-30 paired gain of the false controller (A2 share) over the reading budget q_c (x) and posting budget b (y); base rule. Rows: ρ; columns: q. Tiles are equal size, not to scale. Left of the black line, the vote gate is effectively stricter (3 of 3, 5 of 6 posts).

The phase diagram (Figure 2) separates the three factors that matter:

- **Memory ρ decides** whether the false controller can win at all (top row against bottom row).
- **Reading more defends the proof** under supporting posts: at $\rho = 0.75$ the gain falls from about 0.9 at $q = 3$ to about 0.3 at $q = 12$. Agents who hold pieces of the proof keep posting them, so readers keep meeting them. Under uniform posts this defence disappears (Table 1: 0.67–0.72 at every q).
- **A larger posting budget helps the false controller** ($b = 1 \to 9$: 0.52 → 0.79 at $\rho = 0.75$, averaged over q): each pool is entirely on its target's side, so more posts mean more evidence for A2. The reading budget q_c matters little.

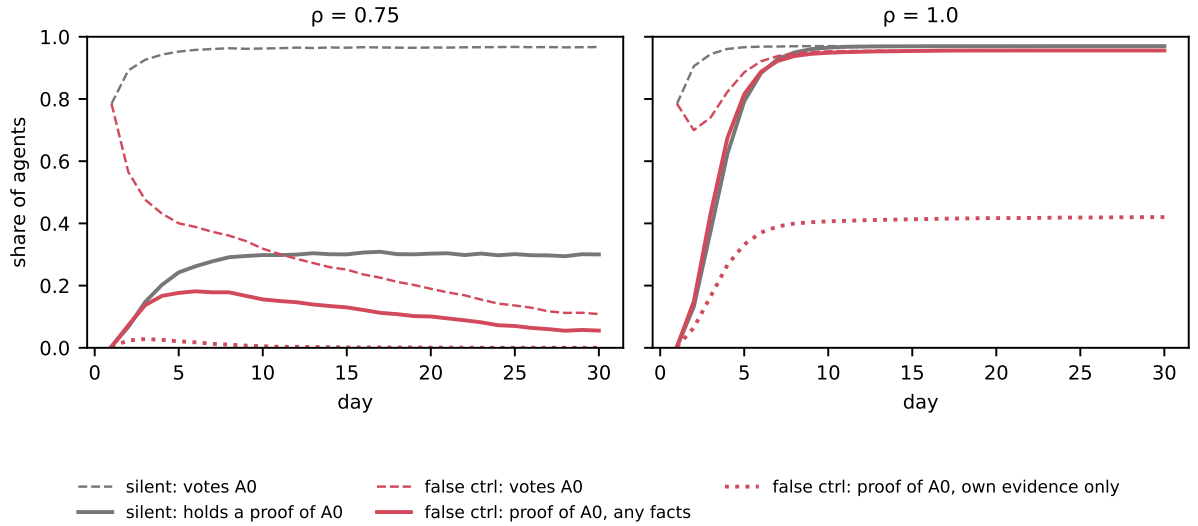


Figure 3: Share of agents voting A0 (dashed) and holding a proof of A0 (solid), silent against false controller. Dotted: proof from the agents' own original evidence only. Base rule, $q = 6$, $q_c = 12$, $b = 3$.

Why memory decides. Figure 3 shows the mechanism. With forgetting, a proof must be reassembled continually; while the false controller fills the board with A2 evidence, the agents' proof decays (own-evidence proofs fall to zero) and the A0 vote follows. With full memory the opposite happens: agents accumulate facts, including the controller's A2 facts. **Those facts are true**, so together with the agents' evidence they complete

proofs of A0: on day 30, 95.6% of agents hold a proof of A0, but only 42% from their own evidence. The rest are proofs the false controller helped build. Every fact a truthful controller posts eventually narrows the possible worlds toward the real one, and with full memory nothing is lost.

3 The controller senses the swarm accurately

Each night the controller estimates the share of agents backing its target from the $n \le q_c$ agent posts it reads:

$$\hat{v} = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{\text{author of post } i \text{ voted for the target}\}, \tag{2}$$

and stays silent when $\hat{v} \ge 0.75$. Each agent posts at most once a day, so the day's board is nearly a census of votes and $\hat{v}$ is an unbiased sample of it.

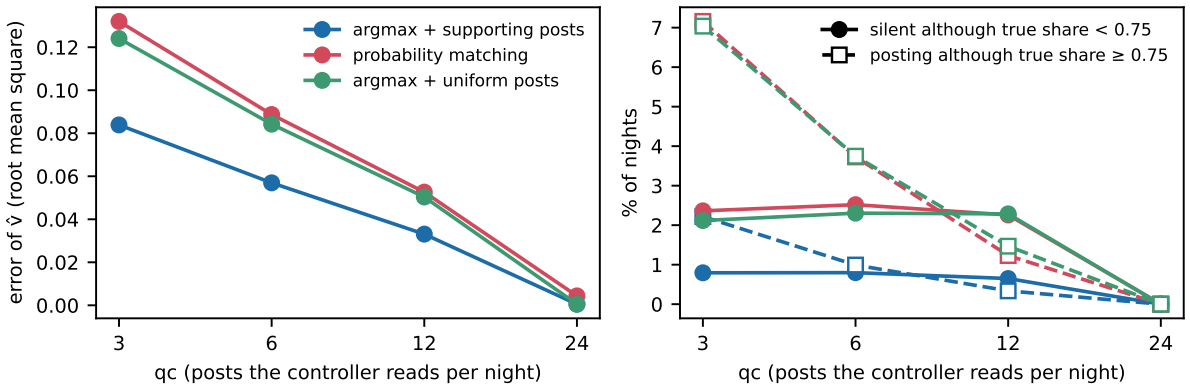


Figure 4: Left: error of $\hat{v}$ against the true share of the 24 agents. Right: gate mistakes. Mean over arms, q, b, ρ; task003-symmetric.

- **No bias**: $\hat{v}$ minus the true share averages within ± 0.0005.
- **The error shrinks with** q_c as random sampling predicts, and vanishes at $q_c = 24$, where the controller reads every post (Figure 4, left).
- **Gate mistakes are rare** (right panel): under 2% of nights at $q_c \ge 12$. They grow at $q_c = 3$, where the gate needs all 3 posts to agree.
- **Abstention does not distort it**: agents almost never abstain (0.1% of agent-days, probability matching only).

So in this setup, **a controller fails because of what it posts, not what it sees.**

4 Helping the truth is not helping agents assemble their own proof

The truth controller stays silent when its sample of the board already proves A0 (`silent_when_target_proved`). We reran all its cells with that rule off and compared, episode by episode, two measures: an agent holds a proof of A0 (i) from any facts it remembers; (ii) from its own original evidence only.

Table 2: Rule off minus rule on, day 30, truth controller, mean over q, q_c, b. "Rule fires": share of nights the controller sees A0 proved; "posts despite proof": share of nights it posts although A0 is proved.

agent rule, ρ	rule fires (on)	posts despite proof (off)	Δ proof, any facts	Δ proof, own evidence
base, 0.75	29%	0.3%	+0.006	−0.012
base, 1.0	32%	0.3%	0.000	−0.029
prob. matching, 0.75	34%	2.3%	+0.016	−0.044
prob. matching, 1.0	31%	0.8%	0.000	−0.057

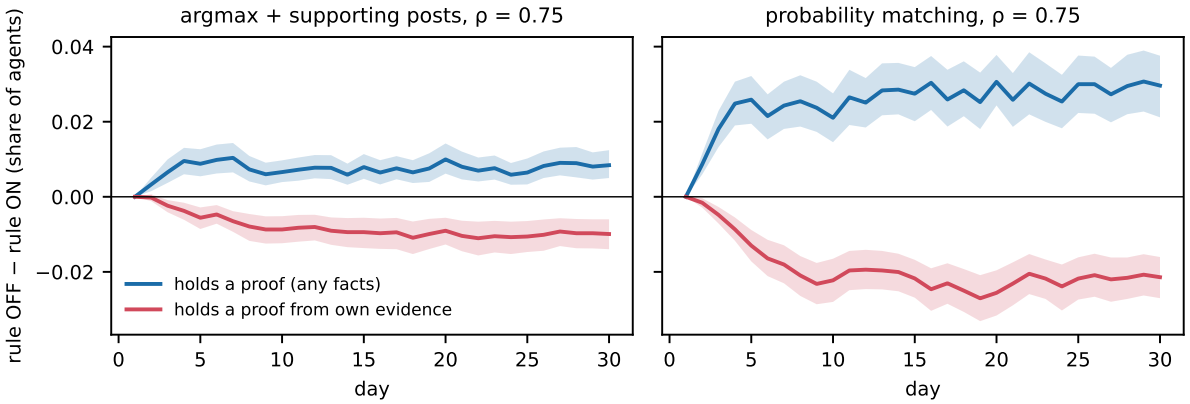


Figure 5: Rule off minus rule on, over days: proof from any facts (blue) and from own evidence (red). $q = 6$, $q_c = 12$, $b = 1$, $\rho = 0.75$.

- **The vote gate already does most of the stopping**: the rule fires on about 30% of nights, yet with it off the controller posts on at most 2% of nights, because by then 75% of the posts it reads already vote A0 (Table 2).
- **The few extra posts pull the two measures apart** (Figure 5): more agents hold a proof, but fewer hold one from their own evidence.
- **Mechanism: posting, not reading.** Controller posts are at most 1.5% of what agents read. But an agent passes on one fact a day, and a controller fact it read on an earlier day now competes with its original evidence for that slot (Table 3). The original evidence circulates less, so fewer agents assemble it into a proof.
- Figure 3 shows the same split for the false controller, much larger: at $\rho = 1$, 95.6% of agents hold a proof of A0 but only 42% from their own evidence.

Table 3: What agents read and post, rule on vs off. Truth controller, probability matching, $\rho = 0.75$, $q = 6$, $q_c = 12$, $b = 3$, all days.

rule	reads from controller	posts of own evidence	posts of pool facts	own facts held
on	1.34%	78.3%	27.3%	5.81
off	1.50%	76.1%	29.3%	5.67

5 How efficiently does the controller steer?

A controller can change the swarm in two ways: in the direction it is asked to, or in ways that happen whatever its target (for example pulling agents out of A1, or splitting them). We call the share of the first kind its *efficiency*. The target is chosen by a fair coin, so how much the swarm's state reveals about it is a well-defined information.

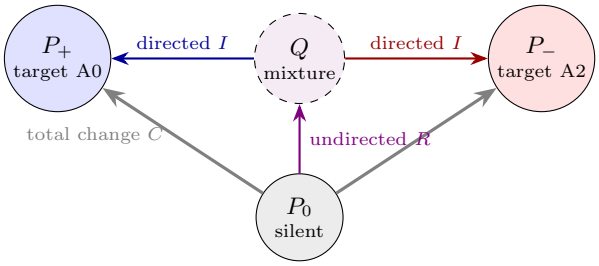


Diagram 2. Three distributions of the swarm's state at time t, one per arm, each over the episodes of that arm. The controller's total change C splits exactly into a part that depends on its target (I, from the mixture Q to each arm) and a part that does not (R, from silence to Q). Efficiency $\eta = I/C$ is the directed share.

For each arm a (silent 0, target A0 +, target A2 −) and a fixed time t, each episode e contributes a probability

vector $y_a(e)$ over the values x of the state X, and

$$\hat{P}_a(x) = \frac{1}{n}\sum_{e=1}^{n} y_a(e)(x), \qquad I = \tfrac{1}{2}\,\mathrm{KL}(\hat{P}_+\|\hat{Q}) + \tfrac{1}{2}\,\mathrm{KL}(\hat{P}_-\|\hat{Q}), \qquad C = \tfrac{1}{2}\,\mathrm{KL}(\hat{P}_+\|\hat{P}_0) + \tfrac{1}{2}\,\mathrm{KL}(\hat{P}_-\|\hat{P}_0), \quad (3)$$

with $\hat{Q} = \frac{1}{2}(\hat{P}_+ + \hat{P}_-)$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$. Exactly $C = I + \mathrm{KL}(\hat{Q}\|\hat{P}_0)$, and $\eta = I/C \in [0,1]$. Two resolutions: **one agent** (X = the vote of a randomly chosen agent; $y_a(e)$ = that episode's vote shares) and the **swarm** ($X = s = m_{\mathrm{A}0} - m_{\mathrm{A}2}$, the difference of the agents' mean beliefs, in three equal bins on $[-1,1]$; $y_a(e)$ marks the episode's bin). At swarm level the controlled swarm often reaches states the silent swarm never visits, which makes C infinite; there we use the bounded version $\eta_{JS} = I/(I + \mathrm{JS}(\hat{Q},\hat{P}_0))$, with JS the Jensen–Shannon divergence. We also measure how unsure the agents become: $\bar{H}$, the mean entropy of the agents' beliefs (0 to 1.58 bits), and the disagreement JSD $= H(\text{mean belief}) - \bar{H}$, each as controlled minus silent in the same episode ($\Delta\bar{H}$, ΔJSD). All quantities are taken at one time, across episodes; they are not divergences between whole trajectories (those would be at least as large). Setting: probability matching, $q = 6$, $q_c = 12$; 1,000 episodes per arm.

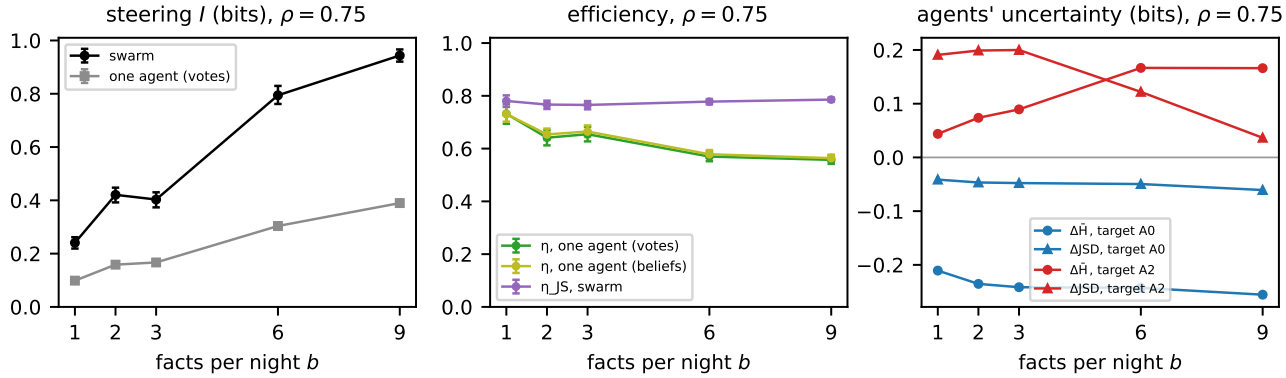


Figure 6: Steering I (left), efficiency (middle) and the change in the agents' uncertainty (right; circles $\Delta\bar{H}$, triangles ΔJSD) at $t = 30$ against facts per night b ($q = 6$, $q_c = 12$). Bars: 95% intervals.

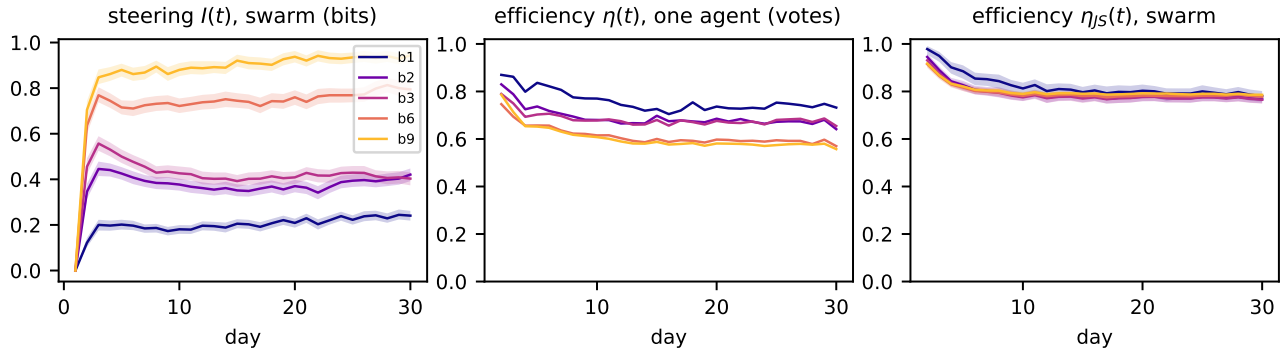


Figure 7: Over time, $\rho = 0.75$, one line per b, end of each day. Bands: 95% intervals.

Findings. (Figures 6, 7; probability matching, $\rho = 0.75$.)

- **Larger batches steer more but less cleanly.** From $b = 1$ to $b = 9$ the swarm reveals the target better (I from 0.24 to 0.94 bits; one agent 0.10 to 0.39), while the one-agent efficiency falls from 0.73 to 0.56: a growing part of the change happens whatever the target.
- **The false controller confuses rather than convinces.** The truth controller makes agents surer ($\Delta\bar{H} = -0.21$ to -0.26 bits) and more alike (ΔJSD -0.04 to -0.06). The false controller makes them less sure ($+0.04$ to $+0.17$) and splits them (ΔJSD $+0.19$ at $b \le 3$).
- **With full memory there is nothing to measure**: every agent ends up proving A0, both controllers leave the same state, $I = 0$ and efficiency is undefined.
- **100 paths are enough for differences of about 0.03 or more** (Table 4); efficiency is the least precise quantity here, because the effects on symmetric are small.

Table 4: Precision. Value from 1,000 paths per arm, and root-mean-square error of the estimate from 50, 100 or 300 paths (200 random subsets each), at $t = 30$, $b = 1$, $\rho = 0.75$. *: share of subsets in which the quantity is infinite or undefined.

quantity	1,000 paths	error, 50	error, 100	error, 300
steering I, swarm (3 bins)	0.241	0.053	0.034	0.018
steering I, one agent (votes)	0.099	0.021	0.013	0.007
efficiency η, one agent (votes)	0.732	0.082	0.058	0.030
efficiency η_{JS}, swarm	0.781	0.055	0.042	0.018
$\Delta\bar{H}$, target A0	-0.211	0.032	0.020	0.010
$\Delta\bar{H}$, target A2	0.044	0.035	0.024	0.011
ΔJSD, target A0	-0.041	0.010	0.007	0.003
ΔJSD, target A2	0.191	0.020	0.014	0.006

Caveats. One world (task_003), 24 agents, exact-Bayesian agents; numbers are means over 1,000 episodes per cell. Full analysis, code and every table: `analysis/task003_llm_free/sim1_analysis/` (reports Q0–Q6).